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General Relativity

Lecture Notes

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Course outline

The course is planned along the following chapters. Chapters that are not yet covered by these notes are listed for orientation only.

1. **Special relativity (SR)**
   - Principles and spacetime diagrams
   - Lorentz transformations
   - Metric
   - Geodesics
2. **SR in arbitrary coordinates** ("gravity" from coordinates)
   - Equivalence principle
   - Γ from coordinate transformations
   - Geodesic equation and metric
   - Metric as gravitational potential: weak-field limit, gravitational redshift
3. **Curvature**
   - Equivalence principle $\Rightarrow$ tides are the true measure of gravity
   - Tides = curvature of spacetime
   - Vectors and tensors: transformation properties
   - Manifolds, differential structure
   - Parallel transport, hence geodesics
   - Geodesic deviation and curvature
   - Connection to the weak-field limit, equivalence principle and gravitational redshift
4. **Vacuum solutions** ("pre" Einstein equations)
   - Schwarzschild solution: geodesics, lensing, horizons
   - Wave solutions: gravitational waves
5. **Matter**
   - Fluids in SR: $T_{\mu\nu}$
   - Fluids in GR: $T_{\mu\nu}$
   - Equation of motion: $\nabla_\mu T^{\mu\nu} = 0$
6. **Einstein equations**
   - $3 + 1$ split
   - Application: FLRW (symmetry arguments)
7. **Cosmology**

1 Special Relativity

LECTURE 01

This chapter is a functional introduction to special relativity (SR), not a historical one: it collects the concepts and tools that are needed later for general relativity (GR).

1.1 The two postulates

Special relativity rests on two fundamental postulates.

(I) **Principle of relativity** (Galileo): no experiment can measure an observer's absolute velocity.

(II) **Constancy of the speed of light** (Einstein): the speed of light is $c = 3{\cdot}10^8\,\mathrm{m\,s^{-1}}$ relative to *any* unaccelerated observer.

Galilean relativity. Newton's laws are invariant under the transformation

$$\vec{v} \;\to\; \vec{v}\,' = \vec{v} + \vec{V} \tag{1.1}$$

for any constant $\vec{V}$, and hence obey postulate (I). The first law says that the velocity stays constant in the absence of forces, which remains true after adding a constant. The second law reads

$$\frac{\vec{F}}{m} = \frac{\mathrm{d}\vec{v}}{\mathrm{d}t} \quad\Longrightarrow\quad \frac{\vec{F}\,'}{m} = \frac{\mathrm{d}\vec{v}\,'}{\mathrm{d}t} = \frac{\mathrm{d}\vec{v}}{\mathrm{d}t} = \frac{\vec{F}}{m}\,, \tag{1.2}$$

so the force is the same for both observers.

Acceleration, in contrast, *can* be detected: imagine a marble lying on the floor of a train, with and without the train accelerating. It is therefore useful to single out the *unaccelerated*, or *inertial*, observers.

Galilean relativity is insufficient. Postulate (II) implies that the Galilean velocity-addition law (1.1) cannot be correct. Otherwise consider $\vec{v}\,' = \vec{v} - \vec{V}$ for a light signal with $|\vec{v}| = c$ and an observer moving with $|\vec{V}| = c$ in the same direction: the observer would measure $|\vec{v}\,'| = 0 \neq c$, in contradiction to (II).

Postulate (II) and its consequences are routinely confirmed in experiments, starting from Michelson–Morley all the way to the LHC.

1.2 Inertial observers and events

The notion of an inertial observer in SR is necessarily *non-local*: an inertial observer is an entire coordinate system

$$(t, \vec{x}) = (t, x, y, z) = (x^0, x^1, x^2, x^3) \equiv x^i, \tag{1.3}$$

which labels *events* $\mathcal{P}$. Being inertial means that three conditions must be satisfied:

1. The distance between the points $\mathcal{P}_1 = (t, x_1, y_1, z_1)$ and $\mathcal{P}_2 = (t, x_2, y_2, z_2)$ is independent of t.
2. The clocks at every point are synchronized and measure t at the same rate.
3. Space is Euclidean, i.e. all of Euclidean geometry is valid.

For the time being we assume the existence of a fictitious universe in which inertial observers exist, i.e. where such a coordinate system can be set up throughout space and at any time, say by using a lattice in space with each lattice location carrying a clock. Then, as recorded by an inertial observer,

- the *position* of an event $\mathcal{P}$ is the (x, y, z) coordinate of the nearest lattice point of that inertial grid;
- the *time* of an event $\mathcal{P}$ is the t measured by the clock *at that* (x, y, z) at the instant of the event.

Thus the definition of an event is local, although the inertial coordinate system itself extends non-locally throughout space and for all time.

Remark

- Instead of saying "throughout space and for all time" it is convenient to say "throughout *spacetime*". This has deep ramifications, eventually leading to GR.
- "Inertial observer" and "inertial reference frame" are used synonymously.

1.3 Natural units

Since $c = 3 \cdot 10^8\,\mathrm{m\,s^{-1}}$ is a constant, it is useful to switch to coordinates in which time is measured in metres rather than seconds: define 1 m of time as the time taken for light to travel 1 m of space. Then

$$c = \frac{1\,\mathrm{m}}{\text{time taken to travel } 1\,\mathrm{m}} = \frac{1\,\mathrm{m}}{1\,\mathrm{m}} = 1\,, \tag{1.4}$$

so that $c = 1$ is dimensionless. These are called *natural units* and are used from here on; velocities v are then dimensionless fractions of the speed of light.

Exercise 1.1

If $c = 1$, write the value of 1 m in seconds and of 1 s in metres.

Exercise 1.2

Work out the unit conversions for acceleration, momentum, force and energy in natural units.

1.4 Spacetime diagrams

A spacetime diagram shows the time coordinate t of an inertial observer on the vertical axis and one spatial coordinate x on the horizontal axis. The trajectory of a particle

through spacetime is its *world line*. In natural units a light ray, $|\vec{v}| = 1$, is a line at 45°; world lines of particles with $|\vec{v}| < 1$ are steeper, and lines with $|\vec{v}| > 1$ are shallower. An accelerated observer has a curved world line (Fig. 1.1).

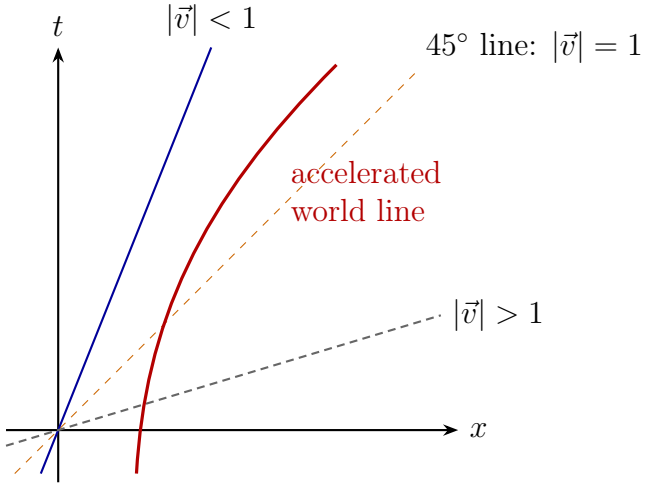


Figure 1.1: Spacetime diagram of an inertial observer in natural units. Light rays are lines at 45°; straight world lines belong to particles of constant velocity.

Exercise 1.3

Use the constancy of c and the definition of an inertial reference frame to construct the $(\bar{t}, \bar{x})$ axes of an observer $\bar{\mathcal{O}}$ in the (t, x) frame of an observer $\mathcal{O}$, where $\bar{\mathcal{O}}$ has relative velocity $v > 0$ in the frame of $\mathcal{O}$ (Schutz, Sect. 1.5). Prove that both axes are tilted by the angle θ with $\tan\theta = |v|$.

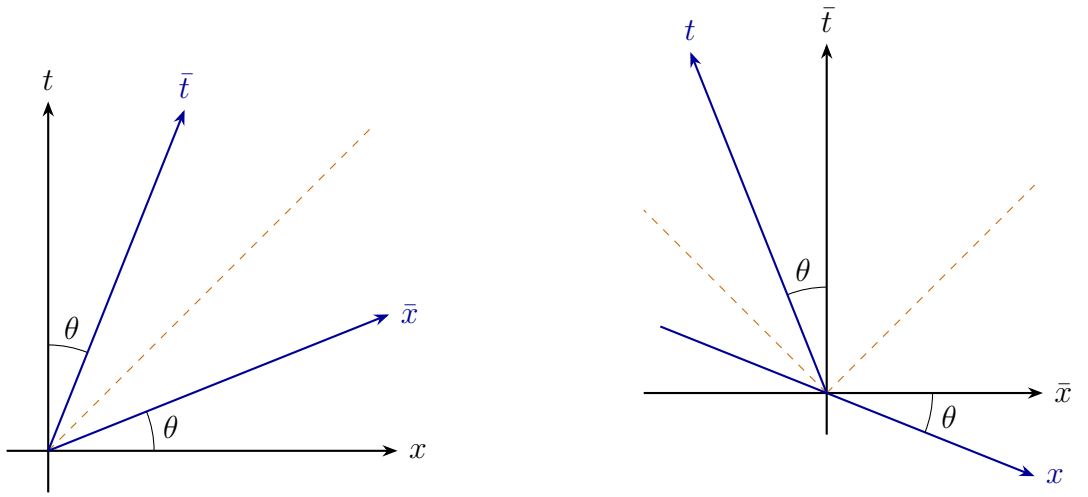
The result of the construction is shown in Fig. 1.2a. The $\bar{t}$ axis is the world line of the spatial origin of $\bar{\mathcal{O}}$, which moves with velocity v in $\mathcal{O}$:

$$\bar{t} \text{ axis } (\bar{x} = 0): \quad x = v\,t\,. \tag{1.5}$$

The $\bar{x}$ axis is the set of events that are simultaneous with the origin for $\bar{\mathcal{O}}$. Using light signals emitted and received on the $\bar{t}$ axis symmetrically around $\bar{t} = 0$ and reflected at such events, the constancy of c gives

$$\bar{x} \text{ axis } (\bar{t} = 0): \quad t = v\,x\,. \tag{1.6}$$

The $\bar{t}$ axis is tilted by θ away from the t axis and the $\bar{x}$ axis by the same angle θ away from the x axis, with $\tan\theta = |v|$; both axes close in symmetrically on the light ray.



(a) Axes of $\bar{\mathcal{O}}$ drawn in the frame of $\mathcal{O}$. **(b)** Axes of $\mathcal{O}$ drawn in the frame of $\bar{\mathcal{O}}$.

Figure 1.2: Coordinate axes of two inertial observers with relative velocity v; dashed lines are light rays. Events on the $\bar{x}$ axis are simultaneous for $\bar{\mathcal{O}}$ but not for $\mathcal{O}$.

Since the $\bar{x}$ axis is not parallel to the x axis, events that are simultaneous for $\bar{\mathcal{O}}$ are not simultaneous for $\mathcal{O}$: we see the **failure of simultaneity**.

Exercise 1.4

Draw the (t, x) axes of $\mathcal{O}$ in the $(\bar{t}, \bar{x})$ frame of $\bar{\mathcal{O}}$. The result is sketched in Fig. 1.2b: $\mathcal{O}$ moves with velocity $-v$ relative to $\bar{\mathcal{O}}$, so its axes are tilted to the other side by the same angle θ.

1.5 Lorentz transformations and invariance of the interval

Consider two observers $\mathcal{O}$ and $\mathcal{O}'$ such that $\mathcal{O}'$ moves with velocity $+v$ relative to $\mathcal{O}$. Without loss of generality we work in $1+1$ dimensions and assume that the origins coincide at the event $\mathcal{A}$. We want to relate the coordinate system (t', x') to (t, x).

Linear ansatz. We assume that the transformation is linear (we return to this assumption later),

$$x^{i'} = \Lambda^{i'}{}_{j}\, x^{j} \qquad \Longleftrightarrow \qquad \begin{pmatrix} x^{0'} \\ x^{1'} \end{pmatrix} = \begin{pmatrix} \Lambda^{0'}{}_{0} & \Lambda^{0'}{}_{1} \\ \Lambda^{1'}{}_{0} & \Lambda^{1'}{}_{1} \end{pmatrix} \begin{pmatrix} x^{0} \\ x^{1} \end{pmatrix}, \tag{1.7}$$

where repeated indices are summed over (Einstein summation convention).

Constraints from the axes. The line $x^{1'} = 0$ is the world line of the observer $\mathcal{O}'$ at the origin, i.e. a straight line through the origin with velocity $+v$ in the frame $\mathcal{O}$, so $x^1 = v\, x^0$:

$$0 = \Lambda^{1'}{}_{0}\, x^0 + \Lambda^{1'}{}_{1}\, v\, x^0 \quad \Longrightarrow \quad \Lambda^{1'}{}_{0} = -v\, \Lambda^{1'}{}_{1}\,. \tag{1.8}$$

The line $x^{0'} = 0$ is the x' axis, which is $x^0 = v\, x^1$ (Exercise 1.3):

$$0 = \Lambda^{0'}{}_{0}\, v\, x^1 + \Lambda^{0'}{}_{1}\, x^1 \quad \Longrightarrow \quad \Lambda^{0'}{}_{1} = -v\, \Lambda^{0'}{}_{0}\,. \tag{1.9}$$

Therefore

$$\Lambda^{i'}{}_{j} = \begin{pmatrix} \Lambda^{0'}{}_{0} & -v\,\Lambda^{0'}{}_{0} \\ -v\,\Lambda^{1'}{}_{1} & \Lambda^{1'}{}_{1} \end{pmatrix} . \tag{1.10}$$

Constraint from light. Now consider an event $\mathcal{B}$ that is connected to $\mathcal{A}$ by a light ray (Fig. 1.3). $\mathcal{B}$ has coordinates $(\Delta t_B, \Delta x_B)$ for $\mathcal{O}$ and $(\Delta t'_B, \Delta x'_B)$ for $\mathcal{O}'$. By postulate (II),

$$\frac{\Delta x_B}{\Delta t_B} = 1 = \frac{\Delta x'_B}{\Delta t'_B} \quad \Longrightarrow \quad -\Delta t_B^2 + \Delta x_B^2 = 0 = -(\Delta t'_B)^2 + (\Delta x'_B)^2 . \tag{1.11}$$

At this stage the combination $-\Delta t^2 + \Delta x^2$ is simply suggestive.

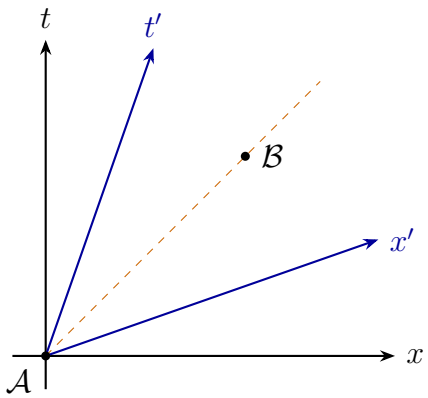


Figure 1.3: The event $\mathcal{B}$ lies on a light ray through $\mathcal{A}$, the common origin of the frames $\mathcal{O}$ and $\mathcal{O}'$.

It is convenient to introduce the matrix η_{ij} with

$$\eta_{00} = -1 , \qquad \eta_{11} = 1 , \qquad \eta_{01} = 0 = \eta_{10} , \tag{1.12}$$

which will later generalise to $1+3$ dimensions. Then, using (1.7),

$$\begin{aligned} 0 &= -(\Delta t'_B)^2 + (\Delta x'_B)^2 = \eta_{i'j'}\,\Delta x'^{i'}_B \Delta x'^{j'}_B = \eta_{i'j'}\,\Lambda^{i'}{}_{i}\Lambda^{j'}{}_{j}\,\Delta x^i_B \Delta x^j_B \\ &= \big[\eta_{i'j'}\Lambda^{i'}{}_{0}\Lambda^{j'}{}_{0}\big]\Delta t_B^2 + 2\big[\eta_{i'j'}\Lambda^{i'}{}_{0}\Lambda^{j'}{}_{1}\big]\Delta t_B \Delta x_B + \big[\eta_{i'j'}\Lambda^{i'}{}_{1}\Lambda^{j'}{}_{1}\big]\Delta x_B^2 . \end{aligned} \tag{1.13}$$

With $\Delta x_B = \Delta t_B$ this becomes

$$0 = \Delta t_B^2 \Big[-(\Lambda^{0'}{}_{0})^2 + (\Lambda^{1'}{}_{0})^2 + 2\big(-\Lambda^{0'}{}_{0}\Lambda^{0'}{}_{1} + \Lambda^{1'}{}_{0}\Lambda^{1'}{}_{1}\big) - (\Lambda^{0'}{}_{1})^2 + (\Lambda^{1'}{}_{1})^2 \Big], \tag{1.14}$$

and inserting (1.10),

$$\begin{aligned} 0 &= -(\Lambda^{0'}{}_{0})^2 + v^2(\Lambda^{1'}{}_{1})^2 + 2\big(v\,(\Lambda^{0'}{}_{0})^2 - v\,(\Lambda^{1'}{}_{1})^2\big) - v^2(\Lambda^{0'}{}_{0})^2 + (\Lambda^{1'}{}_{1})^2 \\ &= \big((\Lambda^{1'}{}_{1})^2 - (\Lambda^{0'}{}_{0})^2\big)\,(1 - 2v + v^2) = \big((\Lambda^{1'}{}_{1})^2 - (\Lambda^{0'}{}_{0})^2\big)\,(1-v)^2 . \end{aligned} \tag{1.15}$$

Since $v \neq 1$, this gives $(\Lambda^{0'}{}_{0})^2 = (\Lambda^{1'}{}_{1})^2$, i.e. $\Lambda^{0'}{}_{0} = \pm\Lambda^{1'}{}_{1}$. We choose the $+$ sign, since $v = 0$ should recover the identity matrix. Defining $\Lambda^{1'}{}_{1} \equiv \gamma(v)$, whose form is unknown for now except for $\gamma(v=0) = 1$, we arrive at

$$\Lambda^{i'}{}_{j} = \gamma(v) \begin{pmatrix} 1 & -v \\ -v & 1 \end{pmatrix} . \tag{1.16}$$

The interval of arbitrary events. Now consider an arbitrary event $\mathcal{C}$ and define the *interval* between $\mathcal{A}$ and $\mathcal{C}$,

$$\Delta s^2 \equiv -\Delta t^2 + \Delta x^2 = \eta_{ij}\,\Delta x^i \Delta x^j\,. \tag{1.17}$$

In the frame $\mathcal{O}'$,

$$\begin{aligned}\Delta s'^2 &= \eta_{i'j'}\,\Delta x^{i'}\Delta x^{j'} = \eta_{i'j'}\,\Lambda^{i'}{}_{i}\Lambda^{j'}{}_{j}\,\Delta x^i \Delta x^j \\ &= \Delta t^2\left[\eta_{i'j'}\Lambda^{i'}{}_{0}\Lambda^{j'}{}_{0}\right] + 2\,\Delta t\,\Delta x\left[\eta_{i'j'}\Lambda^{i'}{}_{0}\Lambda^{j'}{}_{1}\right] + \Delta x^2\left[\eta_{i'j'}\Lambda^{i'}{}_{1}\Lambda^{j'}{}_{1}\right] \\ &= \Delta t^2\left[-\gamma^2 + v^2\gamma^2\right] + 2\,\Delta t\,\Delta x\left[v\gamma^2 - v\gamma^2\right] + \Delta x^2\left[-v^2\gamma^2 + \gamma^2\right] \\ &= \gamma^2(1-v^2)\left(-\Delta t^2 + \Delta x^2\right),\end{aligned} \tag{1.18}$$

so that

$$\Delta s'^2 = \gamma^2(v)\,(1-v^2)\,\Delta s^2\,. \tag{1.19}$$

Fixing γ. Next, consider a frame $\mathcal{O}''$ moving with velocity $-v$ with respect to $\mathcal{O}'$. Clearly $\mathcal{O}'' = \mathcal{O}$. But from (1.19), applied to the pair $(\mathcal{O}', \mathcal{O}'')$,

$$\Delta s''^2 = \gamma^2(-v)\,(1-v^2)\,\Delta s'^2\,. \tag{1.20}$$

Finally, by symmetry, $\gamma(v) = \gamma(-v)$: no spatial direction is preferred, so the intervals between $\mathcal{A}$ and two mirror-image events $\mathcal{C}_{(v)}$ and $\mathcal{C}_{(-v)}$, seen from frames moving with $+v$ and $-v$ respectively, must be equal. Hence

$$\Delta s^2 = \Delta s''^2 = \gamma^2(v)\,(1-v^2)\,\Delta s'^2 = \left[\gamma^2(v)\,(1-v^2)\right]^2 \Delta s^2 \quad\Longrightarrow\quad \gamma^2(v)\,(1-v^2) = 1\,, \tag{1.21}$$

where the positive root is taken because $\gamma^2(1-v^2) > 0$ for $|v| < 1$. Therefore

$$\gamma = \pm\frac{1}{\sqrt{1-v^2}}\,, \tag{1.22}$$

and the $+$ sign is chosen to ensure $t' \to t$ for $v \to 0$.

Invariance of the interval. For arbitrary events,

$$(\Delta s')^2 = (\Delta s)^2\,. \tag{1.23}$$

Two events are called *timelike*, *null* or *spacelike* separated when $\Delta s^2 < 0$, $= 0$ or > 0, respectively.

The Lorentz boost. Collecting the results,

$$\Lambda^{i'}{}_{j} = \gamma\begin{pmatrix} 1 & -v \\ -v & 1 \end{pmatrix}, \qquad \gamma = \frac{1}{\sqrt{1-v^2}}\,, \tag{1.24}$$

so that $x^{0'} = \gamma x^0 - v\gamma x^1$ and $x^{1'} = -v\gamma x^0 + \gamma x^1$, i.e.

$$t' = \frac{t - vx}{\sqrt{1-v^2}}\,, \qquad x' = \frac{x - vt}{\sqrt{1-v^2}}\,, \tag{1.25}$$

which is the same as

$$t = \frac{t' + vx'}{\sqrt{1-v^2}}\,, \qquad x = \frac{x' + vt'}{\sqrt{1-v^2}}\,. \tag{1.26}$$

This transformation is called a **Lorentz boost**.

Generalisation to $1+3$ dimensions. The generalisation is straightforward:

- Choose the velocity direction as the $+x$ axis.
- Apply the boost (1.25) to get (t', x') in terms of (t, x) with $v = |\vec{v}|$.
- Set $y' = y$, $z' = z$.
- This defines $\Lambda^{i'}{}_{j}$ with $i', j = 0, 1, 2, 3$.
- Apply spatial rotations as needed to get any other orientation of the x, y, z axes.

1.6 The metric

By construction,

$$\eta_{ij} = \eta_{i'j'}\,\Lambda^{i'}{}_{i}\Lambda^{j'}{}_{j}\,, \tag{1.27}$$

where η_{ij} and $\eta_{i'j'}$ are the same matrix, in $1+3$ dimensions $\eta_{ij} = \mathrm{diag}(-1, 1, 1, 1)$.

Exercise 1.5

Prove that similarly

$$\eta_{i'j'} = \eta_{ij}\,\Lambda^{i}{}_{i'}\Lambda^{j}{}_{j'}\,, \tag{1.28}$$

where $\Lambda^{i}{}_{j'}$ is the inverse of $\Lambda^{i'}{}_{j}$. Show that the inverse is the boost with the opposite velocity, $\Lambda^{i}{}_{j'}(v) = \Lambda^{i'}{}_{j}(-v)$.

Thus the **metric** η_{ij}, which defines length intervals through

$$\mathrm{d}s^2 = \eta_{ij}\,\mathrm{d}x^i\,\mathrm{d}x^j\,, \tag{1.29}$$

is left invariant by Lorentz boosts. More generally it is invariant under *Poincaré transformations*: 3 boosts and 3 rotations, which together form the *Lorentz transformations*, plus 4 translations in space and time.

Future, past and elsewhere. The interval $\mathrm{d}s^2 = \eta_{ij}\,\mathrm{d}x^i\mathrm{d}x^j$ can be positive, negative or zero. For *each* event $\mathcal{A}$, spacetime is thereby classified into the *future* of $\mathcal{A}$, the *past* of $\mathcal{A}$ and *elsewhere* (Fig. 1.4). The boundary is formed by the light rays through $\mathcal{A}$, the *light cone*. Because the interval is invariant, this classification is the same for all inertial observers.

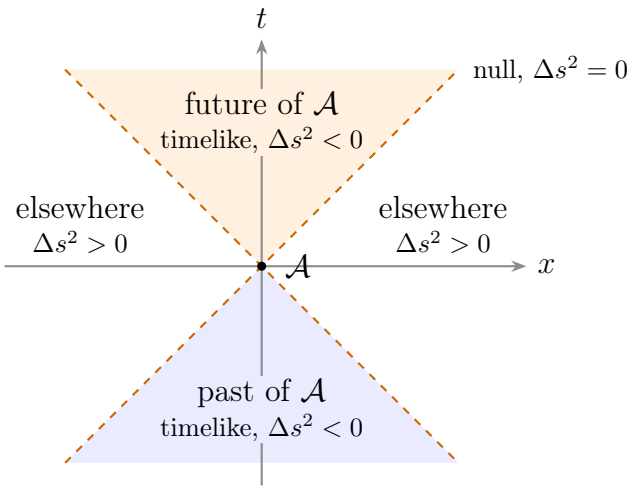


Figure 1.4: Light cone of the event $\mathcal{A}$: future, past and elsewhere.

Exercise 1.6

Draw examples of the future, past and elsewhere for multiple events. Understand the physical meaning of the intersections of these regions.

LECTURE 03

1.7 Invariant hyperbolae

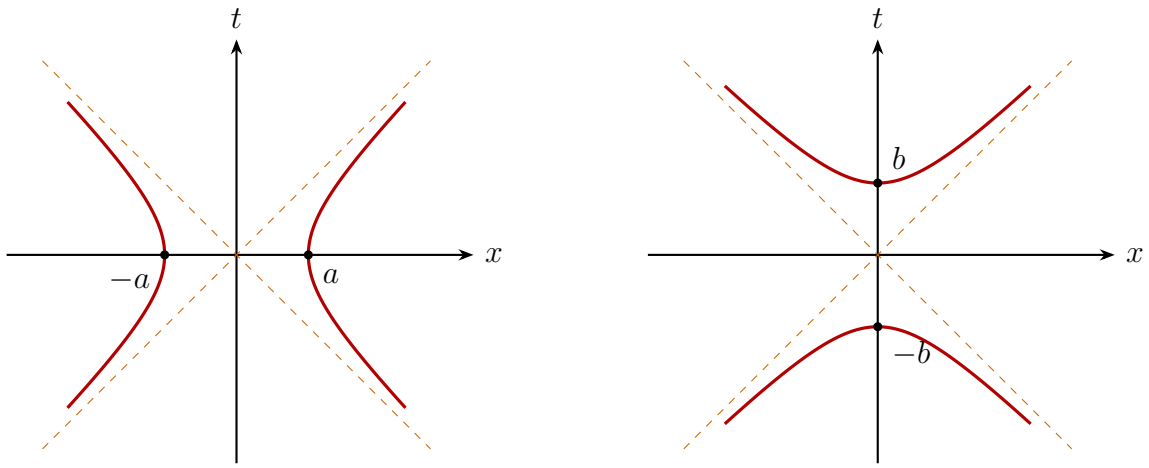
Consider the collection of events (t, x) that have interval a^2 with respect to the origin in the frame $\mathcal{O}$,

$$-t^2 + x^2 = a^2\,, \tag{1.30}$$

which is a hyperbola. But the interval is Lorentz invariant, so in any other frame $\mathcal{O}'$ these events satisfy $-t'^2 + x'^2 = a^2$: the hyperbola is Lorentz invariant. Similarly, the hyperbola

$$-t^2 + x^2 = -b^2 \tag{1.31}$$

is Lorentz invariant (Fig. 1.5). The first family consists of events at spacelike separation from the origin and crosses the x axis at $\pm a$; the second consists of events at timelike separation and crosses the t axis at $\pm b$. Both approach the light cone asymptotically.



(a) $-t^2 + x^2 = a^2$ **(b)** $-t^2 + x^2 = -b^2$

Figure 1.5: The two families of invariant hyperbolae. Every inertial observer assigns the same interval to the events on one branch.

1.8 Time dilation and proper time

Exercise 1.7

Consider events $\mathcal{A}$ and $\mathcal{B}$ on the hyperbola $-t^2+x^2 = -1$ such that $\mathcal{A}$ has coordinates $t = 1$, $x = 0$ (Fig. 1.6). In the frame $\bar{\mathcal{O}}$ in which $\mathcal{B}$ is on the $\bar{t}$ axis, show that $\mathcal{B}$ has $\bar{t} = 1$, $\bar{x} = 0$.

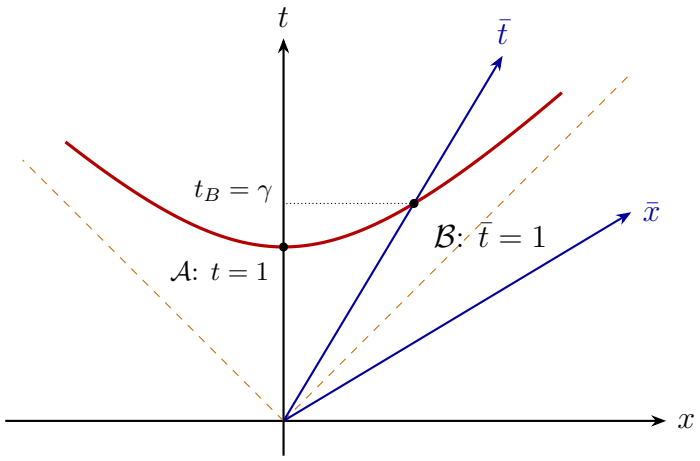


Figure 1.6: Calibrating clocks with the invariant hyperbola $-t^2 + x^2 = -1$: the events $\mathcal{A}$ and $\mathcal{B}$ both lie at unit proper time from the origin, $\mathcal{A}$ on the clock of $\mathcal{O}$ and $\mathcal{B}$ on the clock of $\bar{\mathcal{O}}$.

The exercise shows how clocks can be *calibrated* using invariant hyperbolae: the hyperbola marks, on the time axis of every inertial observer, the event at which that observer's clock shows 1.

It also shows that the clocks in $\mathcal{O}$ and $\bar{\mathcal{O}}$ run at different rates, i.e. they record different times for the same event $\mathcal{B}$, since clearly $t_B > 1$ while $\bar{t}_B = 1$. The line joining $\mathcal{B}$ to the

origin in $\mathcal{O}$ makes an angle $\theta = \tan^{-1}(v)$ with the vertical axis, so $x_B = v\,t_B$ and, from $-t_B^2 + x_B^2 = -1$,

$$t_B = \frac{1}{\sqrt{1-v^2}} = \gamma\,. \tag{1.32}$$

This also follows from setting $\bar{x} = 0$ and $\bar{t} = 1$ in the Lorentz transformation (1.26). In general,

$$(\Delta t)_{\text{measured in } \mathcal{O}} = \gamma \times (\Delta \bar{t})_{\text{measured in } \bar{\mathcal{O}}}\,. \tag{1.33}$$

Here $\Delta\bar{t}$ is the time difference recorded by a *single* clock (the one at rest at $\bar{x} = 0$), while Δt is the time difference between *two* synchronised clocks of $\mathcal{O}$ at different positions. This is **time dilation**: "moving clocks run slower".

Defining the **proper time**

$$\Delta\tau \equiv \sqrt{-\Delta s^2}\,, \tag{1.34}$$

we see that $\Delta\tau$ is the time recorded between two events by a clock in its rest frame. In the example above, $\Delta\bar{t}$ is the proper time of the clock at $\bar{x} = 0$.

1.9 Length contraction

Back to hyperbolae: consider the invariant hyperbola $-t^2 + x^2 = +1$ and take events $\mathcal{A}$ and $\mathcal{B}$ on it such that $\mathcal{A}$ is at $t = 0$, $x = 1$. Consider the frame $\mathcal{O}'$ such that $\mathcal{B}$ is at $t' = 0$. Invariance of the hyperbola implies $x' = 1$ for $\mathcal{B}$, thus calibrating x' in terms of x — rulers are calibrated in the same way as clocks.

Remark

Later we will see that an observer whose world line is such a hyperbola undergoes constant acceleration.

For now, note that the notion of the "length of a rod" is defined using the interval at *fixed time*. Consider a rod at rest in $\mathcal{O}'$, i.e. moving to the right in $\mathcal{O}$, with its rear end passing through the origin (Fig. 1.7). Let $\mathcal{L}_R$ be the rear end at $t = t' = 0$, $\mathcal{L}_F$ the front end at $t = 0$ and $\mathcal{L}'_F$ the front end at $t' = 0$.

- In $\mathcal{O}$, the length of the rod is $\sqrt{\Delta s^2_{L_R L_F}} = \ell_0$.
- In its rest frame $\mathcal{O}'$, the length of the rod is $\sqrt{\Delta s^2_{L_R L'_F}} = \ell$.

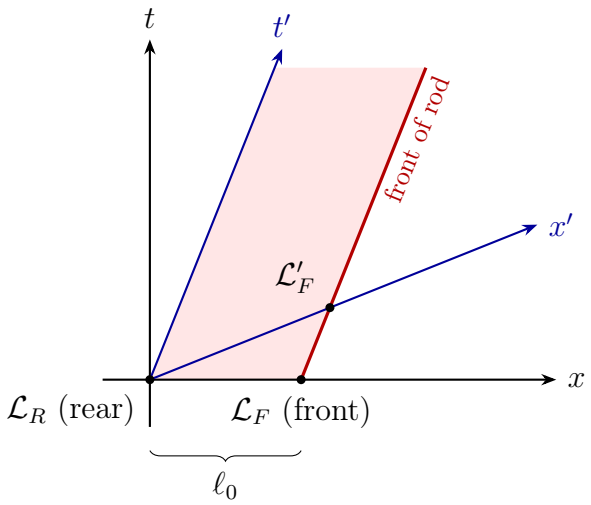


Figure 1.7: A rod at rest in $\mathcal{O}'$. The shaded band is the region of spacetime swept out by the rod. Its length is measured between simultaneous events: $\mathcal{L}_R$ and $\mathcal{L}_F$ for $\mathcal{O}$, but $\mathcal{L}_R$ and $\mathcal{L}'_F$ for $\mathcal{O}'$.

Clearly $\ell_0 < \ell$: this is **Lorentz contraction**. To see by how much, note that the front of the rod has the trajectory $\Delta x/\Delta t = v$, or $x = x_{L_F} + v\,t$. Evaluating this at $\mathcal{L}'_F$ and using the Lorentz transformation (1.26) with $t'_{L'_F} = 0$, $x'_{L'_F} = \ell$,

$$\ell_0 = x_{L_F} = x_{L'_F} - v\,t_{L'_F} = \frac{\ell}{\sqrt{1-v^2}} - \frac{v^2\,\ell}{\sqrt{1-v^2}}\,, \tag{1.35}$$

$$\ell_0 = \ell\,\sqrt{1-v^2} = \frac{\ell}{\gamma}\,. \tag{1.36}$$

These differences (time dilation and length contraction) arise due to a failure in uniquely defining "simultaneous" events. Ultimately this is a consequence of the universality of c. As a result, for example, the notion of rigidity is lost.

Exercise 1.8

Show that the tangent to the hyperbola $-t^2 + x^2 = -b^2$ at any event $\mathcal{P}$ on it is a surface of simultaneity in the frame whose time axis joins $\mathcal{P}$ to the origin.

1.10 Velocity composition

Consider frames $\mathcal{O}$ and $\mathcal{O}'$ such that $\mathcal{O}'$ moves to the right with velocity v in $\mathcal{O}$, and a particle with velocity w' in $\mathcal{O}'$, i.e. $w' = \Delta x'/\Delta t'$. What is the velocity w of the particle in $\mathcal{O}$? From the Lorentz transformation (1.26),

$$\Delta x = \gamma\,(\Delta x' + v\,\Delta t')\,, \qquad \Delta t = \gamma\,(\Delta t' + v\,\Delta x')\,, \tag{1.37}$$

and therefore

$$w = \frac{\Delta x}{\Delta t} = \frac{\Delta t'\,(w'+v)}{\Delta t'\,(1+w'v)}\,. \tag{1.38}$$

$$w = \frac{w' + v}{1 + w'v}\,, \qquad \text{and similarly} \qquad w' = \frac{w - v}{1 - wv}\,. \tag{1.39}$$

Note that $w' \to 1 \Rightarrow w \to 1$ and vice versa. Thus the universality of c is preserved, in contrast to the Galilean law (1.1), which is recovered for $|w'v| \ll 1$.

LECTURE 04

1.11 Geodesics

Geodesics are paths that extremise the path length between two events. The lecture introduces them as the paths that *minimise* path length; in spacetime the extremum is not always a minimum (see the remark at the end of this section). Consider the length of a path between events $\mathcal{A}$ and $\mathcal{B}$ (Fig. 1.8),

$$s_{AB} = \int_A^B \sqrt{|\mathrm{d}s^2|}\,. \tag{1.40}$$

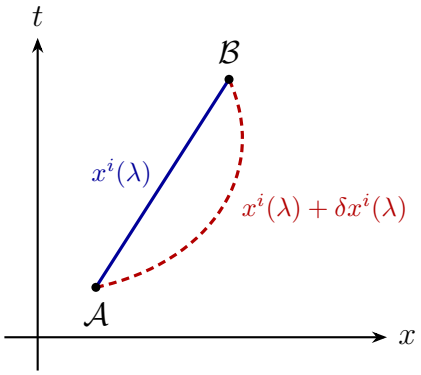


Figure 1.8: Paths between two events $\mathcal{A}$ and $\mathcal{B}$. The path $x^i(\lambda)$ is varied with the end points held fixed.

Choose a parameter λ that increases monotonically along the path. Then

$$s_{AB} = \int_A^B \mathrm{d}\lambda \sqrt{\left|\eta_{ij}\,\frac{\mathrm{d}x^i}{\mathrm{d}\lambda}\frac{\mathrm{d}x^j}{\mathrm{d}\lambda}\right|} = \int_A^B \mathrm{d}\lambda \sqrt{|\eta_{ij}\,\dot{x}^i\dot{x}^j|} \equiv \int_A^B \mathrm{d}\lambda\, L(x^i, \dot{x}^i)\,, \tag{1.41}$$

where $\dot{x}^i \equiv \mathrm{d}x^i/\mathrm{d}\lambda$ is evaluated along the path. Although $L = \sqrt{|\eta_{ij}\,\dot{x}^i\dot{x}^j|}$ depends only on $\dot{x}^i$, for later convenience we assume a generic dependence on x^i as well.

Euler–Lagrange equations. A variation of the path, $x^i(\lambda) \to x^i(\lambda) + \delta x^i(\lambda)$, keeping the end points fixed, gives

$$\delta s_{AB} = \int_A^B \mathrm{d}\lambda\, \delta L = \int_A^B \mathrm{d}\lambda \left[\frac{\partial L}{\partial x^i}\,\delta x^i + \frac{\partial L}{\partial \dot{x}^i}\,\delta \dot{x}^i\right]. \tag{1.42}$$

The variation commutes with the derivative along the path,

$$\frac{\mathrm{d}}{\mathrm{d}\lambda}(\delta x) = \frac{\mathrm{d}}{\mathrm{d}\lambda}\big((x+\delta x)-x\big) = \frac{\mathrm{d}(x+\delta x)}{\mathrm{d}\lambda} - \frac{\mathrm{d}x}{\mathrm{d}\lambda} = \delta\left(\frac{\mathrm{d}x}{\mathrm{d}\lambda}\right), \tag{1.43}$$

so the second term can be integrated by parts; the boundary term vanishes because $\delta x^i = 0$ at $\mathcal{A}$ and $\mathcal{B}$:

$$\delta s_{AB} = \int_A^B \mathrm{d}\lambda \left[\frac{\partial L}{\partial x^i} - \frac{\mathrm{d}}{\mathrm{d}\lambda}\left(\frac{\partial L}{\partial \dot{x}^i}\right)\right] \delta x^i \,. \tag{1.44}$$

Requiring $\delta s_{AB} = 0$ for arbitrary δx^i yields

$$\frac{\partial L}{\partial x^i} - \frac{\mathrm{d}}{\mathrm{d}\lambda}\left(\frac{\partial L}{\partial \dot{x}^i}\right) = 0 \,. \tag{1.45}$$

Working with L^2: affine parameters. Since L contains a square root, it would be easier to work with L^2. We have

$$\begin{aligned}\frac{\partial L^2}{\partial x^i} - \frac{\mathrm{d}}{\mathrm{d}\lambda}\left(\frac{\partial L^2}{\partial \dot{x}^i}\right) &= 2L\,\frac{\partial L}{\partial x^i} - \frac{\mathrm{d}}{\mathrm{d}\lambda}\left[2L\,\frac{\partial L}{\partial \dot{x}^i}\right] \\ &= 2L\left[\frac{\partial L}{\partial x^i} - \frac{\mathrm{d}}{\mathrm{d}\lambda}\left(\frac{\partial L}{\partial \dot{x}^i}\right)\right] - 2\,\frac{\mathrm{d}L}{\mathrm{d}\lambda}\,\frac{\partial L}{\partial \dot{x}^i} \\ &= -2\,\frac{\mathrm{d}L}{\mathrm{d}\lambda}\,\frac{\partial L}{\partial \dot{x}^i}\,,\end{aligned} \tag{1.46}$$

where the last step uses (1.45). But $L = \mathrm{d}s/\mathrm{d}\lambda$. So choosing λ such that s is linear in λ would mean $\mathrm{d}L/\mathrm{d}\lambda = 0$. Such a λ is called an **affine parameter**. For an affine parametrisation,

$$\frac{\partial L^2}{\partial x^i} - \frac{\mathrm{d}}{\mathrm{d}\lambda}\left(\frac{\partial L^2}{\partial \dot{x}^i}\right) = 0 \,. \tag{1.47}$$

For example, for timelike separated events, restricting to variations that keep the path timelike, s_{AB} is just the proper time elapsed along the path. Then we can simply choose $\lambda = \tau$. Similar convenient choices are possible for spacelike and null separated events.

Geodesics of flat spacetime. In the current example, $L^2 = \eta_{ij}\,\dot{x}^i\dot{x}^j$ (up to an overall sign, which drops out of (1.47)), so

$$\frac{\partial L^2}{\partial x^i} = 0\,, \qquad \frac{\partial L^2}{\partial \dot{x}^i} = 2\,\eta_{ij}\,\dot{x}^j \quad \Longrightarrow \quad \eta_{ij}\,\ddot{x}^j = 0\,. \tag{1.48}$$

Introducing the inverse metric $\eta^{ij} = (\eta_{ij})^{-1}$ and contracting with it gives $\ddot{x}^i = 0$:

$$\frac{\mathrm{d}^2 x^i}{\mathrm{d}\lambda^2} = 0 \qquad \text{(geodesics in Cartesian coordinates in SR)}. \tag{1.49}$$

The geodesics of special relativity in Cartesian coordinates are straight lines, $x^i(\lambda) = a^i\lambda + b^i$: exactly the world lines of unaccelerated particles and of light rays.

Remark

For timelike separated events the straight world line is the path of *maximal* proper time among neighbouring timelike paths, not of minimal: any detour makes the elapsed proper time shorter (this is the content of the twin paradox). The condition $\delta s_{AB} = 0$ only selects an extremum, which is all that the derivation above uses.